

Performance Work Statement for the Continual Maintenance of Machine Coolant.

1. **Introduction**: The requiring activity is for Norfolk Naval Shipyard (NNSY), Portsmouth, Virginia. NNSY currently requires a firm, fixed price contract for Continual Maintenance of Machine Coolant.
2. **Performance**: The contractor shall provide all technicians, procurement, handling, management functions, transportation, equipment, parts, tools, materials, supervision and all other items / services necessary to perform on-site continual maintenance of machine coolant for all equipment listed in **Appendix A (Equipment List)**.
3. **Place of Performance**: All services shall be accomplished inside buildings located within the Controlled Industrial Area (CIA) at NNSY (Norfolk Naval Shipyard, Portsmouth, VA 23709). If additional equipment is added to the contract, additional buildings may be added as well. All buildings are less than one mile apart and shall consist of the following:
 - 3.1. 163
 - 3.2. 171
 - 3.3. 202
 - 3.4. 234
 - 3.5. 510
 - 3.6. 1505
4. **Period of Performance**: The Period of Performance will be one Base Year of 12 months and four 12-month option years, including a 6-month option per FAR 52.217-8. The Period of Performance reads as follow:
 - 4.1. Base Year: 4/1/2026 – 3/31/27
 - 4.2. Option Year One: 4/1/2027 – 3/31/28
 - 4.3. Option Year Two: 4/1/2028 – 3/31/29
 - 4.4. Option Year Three: 4/1/2029 – 3/31/30
 - 4.5. Option Year Four: 4/1/2030 – 3/31/31
 - 4.6. Option Year Five: 4/1/2031 – 10/1/31
5. **Equipment Specifications**: Contractor shall perform continual maintenance of machine coolant for all equipment listed in **Appendix A (Equipment List)**.
 - 5.1. During the term of this contract, the Government may remove or replace obsolete or unrepairable equipment or add additional IPE (of similar type and manufacturer) not to exceed 200 pieces of equipment. The inventory list will be updated to address any added or replaced equipment and will be provided to the contractor via contract

modification. Pricing adjustments, if any, will be addressed at time of modification to ensure all equipment to be serviced has full coverage under the contract.

6. Contractor Responsibilities:

6.1. Coolant: Shall be HOCUT 795 manufactured by Houghton International or equivalent.

6.1.1. Contractor shall utilize coolant with all of the following qualities:

- 6.1.1.1.** Resistant to microbial degradation of the fluid
- 6.1.1.2.** Compatible with hard water
- 6.1.1.3.** Bio-stable
- 6.1.1.4.** Reduces the need for antimicrobial agents
- 6.1.1.5.** Low foaming
- 6.1.1.6.** Protects against corrosion
- 6.1.1.7.** Lubricates ways and indexing mechanisms
- 6.1.1.8.** Increases tool life
- 6.1.1.9.** Has a minimum 6-month shelf life

6.1.2. Contractor shall provide knowledgeable and experienced coolant management personnel to fulfill the contract's objectives and duties, including machine-operating skills necessary to drain, circulate and recharge coolant reservoirs.

6.1.3. Contractor shall identify personnel to be trained by Code 106 as Environmental Coordinators (ECs), primary and alternate at no expense to the contractor. The trained ECs will process non-hazardous and hazardous waste as directed by NNSY Code 106 OSHE Program Manual Chapters II-38 and II-45. Copies of these manuals are available upon request.

6.1.4. For sampling, the contractor's ECs shall provide sampling request forms to NNSY Code 106 as directed by NNSY Shop 31. Sampling shall be performed in the bulk container provided by NNSY Code 106. The purpose of this sampling is to ensure the "waste stream" has not changed.

6.1.5. Contractor shall provide 500 gallons (minimum) of premixed coolant to shop personnel for emergent coolant use during peak work operations and weekends.

6.1.6. Contractor will meet with applicable NNSY representative to discuss and resolve coolant maintenance issues on a monthly or as-needed basis.

6.1.7. Contractor shall be responsible for circulating the coolant for one (1) hour bi-weekly on equipment designated by NNSY as low usage (Not to exceed 15% of total machines listed in **Appendix A- Equipment List**).

Note: Circulating coolant is only necessary if HOCUT 795 is not utilized.

6.1.8. Contractor shall provide and own all coolant inventory (utilized coolant for services) and related products.

6.2. Forklift: Forklift operator shall always have a spotter.

6.2.1. Per NNSY OSHE, forklift operation is required a spotter with a whistle while in transits.

6.3. Coolant Cleaning: Any additional machines added to the coolant maintenance program shall be cleaned to meet the following criteria.

6.3.1. Contractor shall remove all coolant, waste liquid and applicable materials as they are generated.

6.3.2. Contractor shall utilize the cleaning procedure in accordance with (IAW) **Section 6.3** of this SOW to remove swarf and debris from machine sumps, return trenches, oil pans, chip conveyors, and filtration units.

6.3.3. Contractor shall use a primary cleaning agent (CIMCLEAN 30 or equivalent) with the following qualities (minimum):

6.3.3.1. Capable of being mixed with water or machine lubricants

6.3.3.2. Inhibits corrosion to machine internals

6.3.3.3. Has anti-bacterial properties

6.3.3.4. Can be used as a concentrate or can be used as a mixture with water

6.3.3.5. Shall be 100% miscible (soluble in water)

6.3.3.6. Able to use as a cleaner when mixed with water

6.3.3.7. Able to be mixed with machine fluids during shutdown in concentrated form

6.3.3.8. Shall be able to remove bio-films, dirt, grit, chip-deposits, insoluble soaps and oils from central systems and sumps

6.3.3.9. Shall be compatible with systems made of ferrous metals

6.3.3.10. Protect machine parts from rusting during the cleaning process

6.3.3.11. Can be used in power washers and steam cleaners

6.3.4. Once clean, contractor shall charge machine coolant system completely with virgin coolant (5 – 7% of the utilized coolant for non-nuclear machinery or 10% for nuclear machinery).

6.3.5. Contractor shall circulate the coolant through the machine system to ensure proper mix and operation.

6.3.6. Contractor shall properly package and label regulated / hazardous waste so that it can be turned over to NNSY Code 106 via the EC for proper disposal. Contractor shall turn in all waste as described in NNSY Code 106 OSHE Program Manual Chapters II-38 and II-45.

- 6.3.7.** Removal of used coolant, cleaning and recharging of new coolant shall be accomplished within eight (8) hours on each machine to prevent rust formation on critical machine components.

Note: The coolant cleaning portion of this service is a one-time requirement for additional machines only. Coolant cleaning shall be accomplished within a four (4) week period, start to finish. The contractor shall coordinate with the shop representative to schedule a machine cleaning and coolant change-out. The shop reserves the right to request a change of the schedule based upon unforeseen workload requirements.

6.4. Coolant Cleaning Procedure:

- 6.4.1.** Remove all swarf and debris from sump; return trenches, oil pans and filtration units.

Note: The term “swarf” is defined as fine chips or filings of stone, metal, or other material produced by a machining operation. i.e. – a curl of metal swarf.

- 6.4.2.** Add one (1) gall of the utilized machine cleaner per 50 gallons of dirty coolant (SHALL BE APPROVED BY NNSY SHOP 31 PRIOR TO USE).
- 6.4.3.** Circulate for four (4) hours through ALL coolant line. Brush all trenches and filter elements, scrub machine, oil pans and exposed surfaces.
- 6.4.4.** When physical cleaning is complete, drain the metal working cleaner from the system.
- 6.4.5.** Fill to capacity with fresh water to remove all traces of cleaner.
- 6.4.6.** REMOVE ALL RINSE WATER.
- 6.4.7.** If rinse water is exceptionally dirty, rinse a second time. Lack of thorough cleaning will cause coolant to become rancid and require re-clean and fresh charge of coolant.

6.5. Coolant Maintenance:

- 6.5.1.** Contractor shall provide services and resources to maintain coolant levels in all equipment listed in **Appendix A (Equipment List)**.
- 6.5.2.** Contractor shall develop and utilize the weekly monitoring procedure IAW **Section 6.5**. This procedure will be applied to each machine on a weekly basis as a minimum.
- 6.5.3.** Contractor shall develop and utilize a clean-out data sheet (CDRL A001) for each machine to include the following:
- 6.5.3.1.** Date service was accomplished
 - 6.5.3.2.** Whether tramp oil was removed and approximate quantity
 - 6.5.3.3.** Initial refractometer reading

6.5.3.4. Whether water or concentrate was added in order to achieve desired coolant concentration

6.5.3.5. Final refractometer reading

6.5.3.6. Service representative signature who performed the work

Note: Contractor shall ensure the monitoring results listed in the data sheet are attached to each machine for inspection purposes

6.5.4. If HOCUT 795 is not utilized as the coolant, contractor shall additionally track coolant performance IAW **Sections 6.6, 6.7, and 6.8.**

6.5.5. Contractor shall remove all waste liquid seven (7) materials generated during the execution of the duties associated with this contract.

6.6. Weekly Monitoring Procedure:

6.6.1. Skim Tramp Oil etc. from coolant

6.6.2. Check refractometer

6.6.3. Non-Nuclear Machinery:

6.6.3.1. If refractometer is above 7% concentration, fill sump with water to capacity

6.6.3.2. If refractometer is below 5% concentration, fill sump from Charge Buggy with 1% premix to capacity (1% premix subject to change)

6.6.4. Nuclear Machinery:

6.6.4.1. If refractometer is above 10% concentration, fill sump with water to capacity

6.6.4.2. If refractometer is below 10% concentration, fill sump from Charge Buggy with 1% premix to capacity (1% premix subject to change)

6.6.5. Re-check refractometer and record readings

6.6.6. If still below 5% (non-nuclear) or 10% (nuclear), add concentrate ONLY

6.6.7. Re-check refractometer and record all readings on data sheet

6.7. Coolant Performance Metrics:

6.7.1. Contractor shall pull a coolant sample for laboratory analysis bi-weekly.

6.7.2. Contractor shall evaluate each laboratory analysis bi-weekly (with the Government) for at least two months, then determine if analysis can go to monthly monitoring.

6.7.3. Contractor shall NOT utilize fungicide to control fungus level(s).

6.7.4. Coolant analysis (CDRL A002) shall include a minimum of the following:

6.7.4.1. Customer

6.7.4.2. NID#

- 6.7.4.3. Product
- 6.7.4.4. Date Charged
- 6.7.4.5. Capacity
- 6.7.4.6. Recommended Contraction %
- 6.7.4.7. Date Sample Taken
- 6.7.4.8. Date Sample Received
- 6.7.4.9. Actual Concentration
- 6.7.4.10. pH Level
- 6.7.4.11. Tramp Oil
- 6.7.4.12. Bacteria Level
- 6.7.4.13. Fungi Level
- 6.7.4.14. Running Charge of Contraction & pH Level
- 6.7.4.15. Recommendations

6.8. Coolant Recycling Criteria:

- 6.8.1. The following details the recycling criteria for Bacteria and Fungi.
- 6.8.2. Desired concentration range is 5-7% for non-nuclear machinery and 10% for nuclear machinery.
- 6.8.3. When the coolant recycling standards fall out of the specified acceptable range the contractor shall remove the old coolant and provide replacement coolant or recycle the old coolant using the coolant recovery machine located in NNSY Shop 31.

Note: Contractor shall be responsible for the maintenance and upkeep of the coolant recovery system

Bacteria

Problem	Action	Recycle Criteria
10,000 biweekly	Monitor closely; Add concentration if needed	None
10,000,000 biweekly	Add concentration to return to desired concentration range	If 2 consecutive biweekly readings occur and the concentration within the desired concentration range

Fungi

Problem	Action	Recycle Criteria
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100 biweekly	Monitor closely; Add concentration if needed to return to desired concentration range	None
100 biweekly	Add concentration to return to desired concentration range	If 2 consecutive biweekly readings occur and the concentration is within the desired concentration range

6.9. Coolant Recharging Criteria:

6.9.1. The following details the recharging criteria for Bacteria and Fungi.

6.9.2. Desired concentration range is 5-7% for non-nuclear machinery and 10% for nuclear machinery.

Bacteria

Problem	Action	Recycle Criteria
10,000,000 biweekly	Add concentration to return to desired concentration range	If 3 consecutive biweekly readings occur and the concentration within the desired concentration range

Fungi

Problem	Action	Recycle Criteria
100 biweekly	Add concentration to return to desired concentration range	If 3 consecutive biweekly readings occur and the concentration is within the desired concentration range

7. **Quality Assurance (QA):** The Government shall evaluate the contractor's performance under this contract IAW the Quality Assurance Surveillance Plan (QASP). This QASP (see **Appendix B - Definitions**) is primarily focused on what the Government shall do to ensure that the contractor has performed IAW the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and acceptable quality level(s) (defect rate(s)).

8. **Shipyard Responsibilities:** The shipyard shall be responsible for the following.

8.1. NNSY shop 31 will monitor the need for waste containers. The Shipyard will provide bulk containers as well as non-bulk Performance Oriented Packaging (POP) as approved by the Government for collection, storage, and disposal of non-bulk hazardous waste if required.

8.2. NNSY will provide familiarization training on the machine coolant systems.

8.3. NNSY will provide the contractor mutually agreed upon lockable storage space for coolant / supplies on the first floor of NNSY Building 171 adjacent to the coolant recovery system and a satellite accumulation area for the hazardous waste is applicable. Contractor shall be responsible for maintaining these areas in a neat and orderly fashion.

8.4. NNSY will notify the contractor within 24 hours of contaminating coolant with phenolic, rubber, etc. to request coolant change-out. (Not to exceed six (6) requests or 200 gallons per year).

9. **Certification of Work Requirements:** The COR shall verify that quality of services are accomplished IAW **Section 7** and **Appendix B - Definitions** of this PWS after each service is complete.

10. **Safety or Security:** Contractors shall abide by the rules of NAVSHIPDNOR M-11450.4B Material Handling Operations for forklift safety operations. Also, contractors shall secure access to the CIA and provide their own Personal Protection Equipment (PPE). “Per FAR 22.102-1(a) entitled ‘Safety’, Contractors are hereby informed that their employees or representatives present on property under Norfolk Naval Shipyard cognizance shall follow Federal and State OSHA requirements; Norfolk Naval Shipyard OSHE Program Manual, NAVSHIPYDNORINST P5090.2 requirements; OSHE Control Manual (OSHECM) requirements; or safety requirements for site specific location of the contract.”

11. Facility Entry Requirements:

11.1. **Initial Base Entry:** 30 days before arrival, contractor shall provide SECNAV 5512 data on all personnel expected to visit NNSY. Personnel visiting NNSY shall be U.S. citizens. The TPOC shall provide SECNAV 5512 forms upon request. The TPOC will endorse the SECNAV 5512 form and forward to the NNSY Pass Office. Personnel having existing valid base passes will be exempt from this requirement. Existing base passes shall be valid for the duration of the expected visit.

11.2. **Vehicle Entry:** The company name shall be displayed on each door of the Contractor’s vehicles in a manner and size that is clearly visible. All vehicles shall display a valid state license plate that complies with State Vehicle Code. Vehicles shall meet all other requirements of the State Vehicle Code, such as safety standards, and shall carry proof of insurance and state registration. Insurance and registration shall be in the company’s name. Passes will be issued for vehicles to enter the shipyard. A separate pass is required for entering the CIA.

11.3. **Defense Biometric Identification System and NAVSEA Badging (REQUIRED FOR ALL CONTRACTS)**

All contractors, with the exception of permanent party (or “seated” contractors with Common Access Cards (CACs), will be required to obtain a DBIDS card to gain access to NNSY.

Procedure for receiving a DBIDS card:

- a. Contractor will complete the SECNAV 5512 form(s) and forward them to their NNSY Sponsor/Contracting Officer's Representative (COR).
- b. The sponsor will complete the CVS spreadsheet utilizing the information on the completed SECNAV5512 forms and forward all documents to Commander Navy region Mid-Atlantic (CNRMA) for vetting.
- c. CNRMA will add the contractor to the "179-Day List" which is utilized by the Pass and ID office to create the DBIDS card.
- d. Once on the 179-Day List, the contractor may go to Pass and ID to have their DBIDS card generated.
- e. Green NAVSEA access badges can be generated at this point with contractors providing proof of U.S. Citizenship (Birth Certificate/ US Passport).

11.4. Portable Electronic Devices (PEDs)

Portable Electronic Device (PED): IAW CONSUBLANT/COMSUBPACINST 2075.1C, the use of PEDs while in or on Department of Navy Controlled Industrial Areas are prohibited. A PED is defined as any non-stationary electronic apparatus with singular or multiple capabilities of, but not limited to recording, storing, and/or transmitting data, voice, video, or photo images e.g.: cell phones, laptops, tablets, and wearable devices such as fitness bands and smart watches. Non-camera cell phones are acceptable (feature phones). Where a laptop computer is required for trouble-shooting and machine programming purposes, contractor shall inform TPOC 2 weeks prior to visit to allow time to schedule inspections and vetting from NNSY IT department.

A PED is defined as any non-stationary electronic apparatus with singular or multiple capabilities of, but not limited to recording, storing, and/or transmitting data, voice, video, or photo images (e.g., cell phones, laptops, tablets and wearable devices such as fitness bands and smart watches).

- a. NAVSEA 04 Ser 04/281 IMPLEMENTATION PLAN FOR ALLOWING CAMERA-PORTABLE ELECTRONIC DEVICES WITHIN THE CONTROLLED INDUSTRIAL AREA
- b. NAVSEAINST 2200.1A PORTABLE ELECTRONIC DEVICES POLICY
- c. SECROMAN Vol 1 5510.36 Rev E

In accordance with ref (a), personally owned camera enabled PEDs will be authorized inside of the Controlled Industrial Area (CIA) and all.

Security Islands. Camera enabled PEDs will remain unauthorized within Controlled Nuclear Information Areas (CNIAs), Nuclear Work Areas (NWAs) and Open Storage Areas.

All PEDs, camera enabled or not, are still prohibited within NWAs and Open Storage Areas.

The allowance of camera enabled PEDs inside of the CIA and Security Islands does not negate existing photography policies. Unofficial photography, filming of video, or live streaming from the camera-enabled PED remains prohibited within the CIA or any NAVSEA controlled space outside of the CIA. Further, audio recordings (both digitally or magnetically) in NAVSEA spaces is prohibited without the approval of the Code 1120 Security Director.

12. Critical Time Constraints: The continual maintenance of machine coolant shall be performed between the hours of 7:20 am and 3:50 pm excluding any federal holidays. Time periods outside of these boundaries may be negotiated on a case-by-case basis.

13. Technical POC:

TBA

14. Contracting Officer Representative (COR):

TBA